A note on the sextic gap polynomial for $N = 3$ Type-1 matrices

Semiformal research note

Context. Numerical elimination suggests that for $N = 3$ Type-1 matrices, each pairwise gap $\Delta_{ij}(t) = E_i(t) - E_j(t)$ satisfies a surprisingly sparse degree-6 polynomial in t and Δ, with only 16 of the 28 a priori monomials present. The purpose of this note is to explain that sparsity analytically, give a compact closed-form factorization, and then assess what this does and does not simplify in the exact-WKB/Stokes picture.

Plan: three high-conviction tactical paths

I see three natural routes.

1. **Universal invariant route.** Ignore Type-1 structure at first and treat any 3×3 linear pencil $H(t) = T + tV$. Form the monic sextic

$$\Phi(t, \Delta) = \prod_{1 \le i < j \le 3} \left(\Delta^2 - (E_i - E_j)^2 \right).$$

   If one can express its coefficients through trace invariants of $H(t)$, then the observed sparsity should become automatic rather than miraculous.

2. **Resolvent / depressed-cubic route.** The cubic characteristic polynomial has a standard resolvent. If the three squared gaps can be written as simple functions of one depressed-cubic variable, the pairwise sector may reduce to an algebraic curve of very low complexity.

3. **Type-1 specialization.** Feed the Type-1 parametrization

$$u(x) = \sum_{m=1}^{3} \frac{\gamma_m^2}{x - \epsilon_m}, \qquad E(x) = \sum_{m=1}^{3} \frac{d_m \gamma_m^2}{x - \epsilon_m}$$

   into the invariant formulas and check whether the resulting pairwise exact-WKB differential becomes rational on the known x-sphere. If yes, the ordinary pairwise Stokes sector should simplify sharply, even if virtual turning points remain.

All three routes succeed, though the third only partially closes the full exact-WKB story.

1. Universal derivation of the sextic

Let e_1, e_2, e_3 denote the three eigenvalues of a generic 3×3 matrix H and define

$$y_{12} = (e_1 - e_2)^2, \qquad y_{13} = (e_1 - e_3)^2, \qquad y_{23} = (e_2 - e_3)^2.$$

The natural monic cubic in the variable $y = \Delta^2$ is

$$P_H(y) = \prod_{i<j} (y - y_{ij}) = y^3 - \sigma_1 y^2 + \sigma_2 y - \sigma_3.$$

Let

$$s_1 = e_1 + e_2 + e_3, \qquad s_2 = e_1e_2 + e_1e_3 + e_2e_3, \qquad s_3 = e_1e_2e_3.$$

A direct symmetric-polynomial calculation gives

$$\sigma_1 = 2(s_1^2 - 3s_2), \qquad \sigma_2 = (s_1^2 - 3s_2)^2, \qquad \sigma_3 = \prod_{i<j}(e_i - e_j)^2.$$

Introduce the two standard cubic invariants

$$A := s_1^2 - 3s_2, \qquad D := \prod_{i<j}(e_i - e_j)^2 = \mathrm{Disc}_\lambda(\lambda^3 - s_1\lambda^2 + s_2\lambda - s_3).$$

Then

$$\boxed{P_H(y) = y^3 - 2A\,y^2 + A^2\,y - D,} \qquad y = \Delta^2.$$

Equivalently,

$$\boxed{\Phi_H(\Delta) = \Delta^6 - 2A\,\Delta^4 + A^2\,\Delta^2 - D.}$$

This is the closed-form explanation for the numerical sparsity.

Trace form

For a 3×3 matrix H one may rewrite

$$A = \frac{1}{2}(3\ \mathrm{tr}\,H^2 - (\mathrm{tr}\,H)^2),$$

and

$$D = \frac{4A^3 - B^2}{27}, \qquad B := 2(\mathrm{tr}\,H)^3 - 9(\mathrm{tr}\,H)(\mathrm{tr}\,H^2) + 9\ \mathrm{tr}\,H^3.$$

So for any linear pencil $H(t) = T + tV$,

$$A(t) \text{ is quadratic in } t, \qquad B(t) \text{ is cubic in } t, \qquad D(t) \text{ is sextic in } t.$$

Therefore

$$\Phi(t, \Delta) = \Delta^6 - 2A(t)\Delta^4 + A(t)^2\Delta^2 - D(t)$$

contains at most

$$1 + 3 + 5 + 7 = 16$$

monomials:

$$\Delta^6, \quad t^0, t^1, t^2 \text{ with } \Delta^4, \quad t^0, \ldots, t^4 \text{ with } \Delta^2, \quad t^0, \ldots, t^6 \text{ with } \Delta^0.$$

So the "12 out of 28 coefficients vanish identically" observation is not an empirical accident; it is a theorem.

2. Resolvent form and a second factorization

Define

$$M_k := s_1 - 3e_k, \qquad k = 1, 2, 3.$$

Substituting $e_k = (s_1 - M_k)/3$ into the characteristic equation shows that the M_k are roots of the depressed cubic

$$\boxed{M^3 - 3A\,M + B = 0.}$$

For the pair complementary to e_k (for example $k = 3$ corresponds to the pair $(1, 2)$), one finds

$$\boxed{\Delta_{ij}^2 = \frac{4A - M_k^2}{3}.}$$

This is already a strong simplification: every pairwise squared gap is a simple algebraic function of one resolvent branch.

Eliminating M gives a second closed form:

$$\boxed{B^2 = (4A - 3\Delta^2)(3\Delta^2 - A)^2.}$$

This is equivalent to the sextic above, because

$$\Delta^2(\Delta^2 - A)^2 = D = \frac{4A^3 - B^2}{27}.$$

3. Type-1 specialization

Now let $H(t)$ be a Type-1 matrix. We have the explicit spectral parametrization

$$t = u(x) = \sum_{m=1}^{3} \frac{\gamma_m^2}{x - \epsilon_m}, \qquad E(x) = \sum_{m=1}^{3} \frac{d_m \gamma_m^2}{x - \epsilon_m}.$$

Choose the unique quadratic polynomial

$$f(z) = az^2 + bz + c \qquad \text{with} \qquad f(\epsilon_m) = d_m.$$

Then one may write

$$E(x) = f(x)\,u(x) - \ell(x),$$

where $\ell(x)$ is linear. On two sheets $x_i(t), x_j(t)$ this gives

$$\Delta_{ij}(t) = E(x_i) - E(x_j) = (x_i - x_j)\Big[t\big(a(x_i + x_j) + b\big) - a\Gamma_0\Big], \qquad \Gamma_0 = \sum_{m=1}^{3} \gamma_m^2.$$

This is the factorization we had already isolated numerically. The new point is that it is now subsumed by the universal invariant identity above.

Consequence for the ordinary pairwise sector

Let $y = \Delta_{ij}^2$. Then y is one of the three roots of

$$y^3 - 2A(t)y^2 + A(t)^2 y - D(t) = 0.$$

Hence the ordinary pairwise WKB differential

$$\lambda_{ij} = \Delta_{ij}(t)\, dt$$

lives on the algebraic pair curve

$$\boxed{y^3 - 2A(t)y^2 + A(t)^2 y - D(t) = 0.}$$

Equivalently, using the resolvent variable M,

$$\lambda_{ij} = \pm\sqrt{\frac{4A(t) - M^2}{3}}\, dt, \qquad M^3 - 3A(t)M + B(t) = 0.$$

For Type-1, $A(t)$ and $B(t)$ are not abstract: they come from the explicit x-sphere parametrization, so the pair curve is birational to a rational cover of the x-sphere. This is a real simplification of the *ordinary* pairwise Stokes problem.

4. What simplifies in the Stokes picture

This structure buys us a lot.

- **Ordinary turning points.** They are now simply the zeros of $D(t)$, i.e. the discriminant locus of the cubic characteristic polynomial. This is immediate from

$$D(t) = \prod_{i<j}(E_i - E_j)^2.$$

- **Ordinary Stokes curves.** For each pair (i, j) they are critical trajectories of the quadratic differential

$$q_{ij}(t) = \Delta_{ij}(t)^2\, dt^2 = y(t)\, dt^2$$

  on the pair curve. Since y is algebraic of degree three over the t-plane, the ordinary sector is a low-complexity algebraic problem rather than an opaque three-sheet one.

- **Pairwise exact-WKB actions.** They reduce to Abelian integrals of the explicit 1-form

$$\lambda_{ij} = \pm\sqrt{y(t)}\, dt$$

  or, on the resolvent cover,

$$\lambda_{ij} = \pm\sqrt{\frac{4A(t) - M^2}{3}}\, dt.$$

  So one no longer needs to think of the pairwise sectors as arbitrary multi-sheet objects.

In this sense the new polynomial structure does provide a simplification comparable to the earlier x-cover reduction.

5. What this does *not* trivialize

It does *not* trivialize the genuinely higher-order part of the exact-WKB problem.

Virtual turning points and new Stokes curves are created by *interactions between different pair sectors*. The data one still has to compare are, schematically,

$$\int \lambda_{12}, \qquad \int \lambda_{23}, \qquad \int \lambda_{13},$$

together with the half-form/sign data and the logarithmic monodromy of the Type-1 primitive. So the pairwise algebraic simplification solves the ordinary local geometry but not yet the global selector.

What it *does* do is sharpen the remaining problem:

> The unresolved exact-WKB content is now the gluing of three explicit algebraic pairwise sectors, rather than the analysis of an amorphous rank-3 Stokes graph from scratch.

6. Successes and challenges

Successes.

- The sextic gap polynomial is now derived in closed form.
- The observed monomial sparsity is completely explained.
- The coefficients are controlled by only two scalar invariants, $A(t)$ and $D(t)$ (or $A(t)$ and $B(t)$).
- The ordinary pairwise Stokes sector reduces to algebraic quadratic differentials on an explicit low-complexity cover.

Challenges.

- The simplification does not by itself identify virtual turning points.
- The selector still has to compare actions across *different* pair curves.
- Closed-form fine chamber walls still appear to require a global exact-WKB / continuation analysis, not just one-pair algebra.

7. Numerical sanity check

I verified the identity

$$\Delta^6 - 2A(t)\Delta^4 + A(t)^2\Delta^2 - D(t) = 0$$

on 100 random real Type-1 samples. The maximum absolute residual over all tested pairwise gaps was

$$5.82 \times 10^{-9},$$

with median residual

$$8.42 \times 10^{-13}.$$

(Residuals are purely numerical; the derivation above is exact.)

Bottom line

The experimental sextic is not merely a curiosity. For $N = 3$, it is the natural and universal gap polynomial

$$\boxed{\Delta^6 - 2A(t)\Delta^4 + A(t)^2\Delta^2 - D(t) = 0,}$$

with

$$A(t) = \frac{1}{2}\big(3\,\mathrm{tr}\,H(t)^2 - (\mathrm{tr}\,H(t))^2\big), \qquad D(t) = \mathrm{Disc}_\lambda\,\chi_{H(t)}(\lambda).$$

For Type-1 matrices this means the ordinary pairwise exact-WKB sector is substantially simpler than it first appeared: it is controlled by explicit algebraic differentials on a rationally parametrized cover. What remains nontrivial—and probably genuinely interesting—is the higher-order gluing of those three pairwise sectors, i.e. the virtual turning points, new Stokes curves, and the global selector.

My current highest-conviction next move. Recast the virtual-turning-point / selector problem entirely in the resolvent variables (A, B, M) and ask whether the equal-action cover can be written as a finite correspondence between the three algebraic pair curves. If that works, the fine exact-WKB chamber problem may collapse to a finite gluing problem among three explicitly known Abelian differentials.